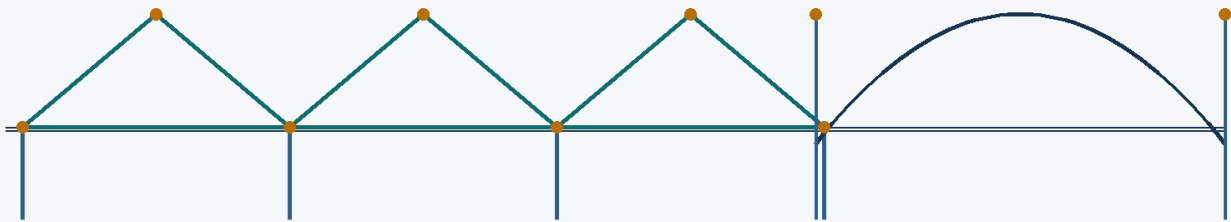


CE 3111

STRUCTURAL ANALYSIS AND DESIGN - I

QUESTION BANK ANALYSIS

Topic-wise Set Distribution System



8 SETS | 4 PER SECTION | ANSWER 3 OF 4 FROM EACH SECTION

SOURCE WINDOW	Regular QB 2016-2024 + 2024 Backlog
ANALYSIS BASIS	Question Bank + class notes + lecture slides + solved QB files
CREDIT	Made by 2223033

Built for selective preparation: master 3 high-confidence sets and keep 1 backup in each section.

How this analysis was built

Important section-order note Historical papers do not keep the same printed Section A/B order. The 2024 Regular paper and the 2024 Backlog paper contain essentially the same eight set families but print the two section blocks in opposite order. For consistency, this guide uses the 2024 Regular set order; always identify a set by topic, not only by the letter printed on an old paper.

- Primary corpus: complete regular examination papers from 2016-2019 and 2021-2024 (8 complete full-length papers).
- 2020 is a shortened/partially scanned paper in the uploaded QB; it is used as supporting evidence but excluded from the 8/8 complete-paper hit rate.
- 2024 Backlog is used as a confirmation paper and as the latest alternative paper pattern.
- A topic is counted as the same question pattern when the geometry/data change but the required method is essentially the same.
- The question wording below is paraphrased for fast revision; marks and exact figures should be checked from the original QB when practicing.

What "ANNUAL CORE" means

The set family is present in all 8 complete full-length regular papers reviewed. The exact pairing of subtopics inside a 35-mark question changes from year to year, but the family itself is highly persistent.

Exam rule that drives the strategy

3-of-4 rule You need to answer 3 sets from each section out of 4. Therefore, the safest preparation model is not "study everything equally"; it is "master 3 annual-core sets + prepare 1 backup set well enough to switch if the paper is unusual."

Eight-set distribution at a glance

SECTION A Core analysis, loads and deflection

Set	Topic family	Frequency	Typical character
SET 1	Stability & Determinacy + SFD/BMD	Annual Core	Variable classification + diagram set
SET 2	Truss + Three-Hinged Arch	Annual Core	Member force + arch force/moment
SET 3	Wind + Earthquake	Annual Core	Most algorithmic / data-driven set
SET 4	Deflection: Unit Load/Virtual Work + Conjugate Beam	Annual Core	Repeated theory/proof + numerical

SECTION B Moving load, approximate analysis and indeterminate systems

Set	Topic family	Frequency	Typical character
SET 5	Influence Line Diagram (ILD)	Annual Core	Beam + frame ILD; theory appears too
SET 6	Approximate Frame Analysis: Portal + Cantilever + Factor	Annual Core	Recent trend: portal + cantilever
SET 7	Mill Bent / Portal Application + Space Truss	Annual Core	Space truss is the stable core; mill bent recurs
SET 8	Three-Moment Equation + Suspension Cable	Annual Core	Continuous beam + cable tension/dip/length

Recommended 3-of-4 preparation route

Section	Default 3-set route	Backup / switch option
A	Set 3 + Set 4 + Set 2	Set 1 if you are stronger in stability/SFD than truss/arch
B	Set 5 + Set 8 + Set 6	Set 7 if you are stronger in 3D equilibrium / mill-bent analysis

Why this route? This is a study recommendation based on repetition plus method predictability, not a guarantee of difficulty. If your personal strength is different, swap the backup set in.

SECTION A**| SET 1****STABILITY & DETERMINACY + SFD/BMD****ANNUAL CORE****8/8 complete regular papers | 2024 backlog confirms****What this set usually contains**

- Classify beams, frames, trusses and arches as stable/unstable and determinate/indeterminate.
- Find reactions and draw axial-force, shear-force and bending-moment diagrams of determinate frames/beams.
- Short theory often appears before the numerical: differences, zero-force member, or classification rules.

Most repeated patterns - highest priority first

- | | |
|-----------|--|
| #1 | Structure classification - Stable/unstable + statically determinate/indeterminate is the single most repeated pattern in this set. |
| #2 | Frame reactions + SFD/BMD - A loaded rigid-jointed frame is given; find reactions and draw shear/bending diagrams, sometimes also axial-force diagrams. |
| #3 | Difference questions - Differentiate determinate vs indeterminate and stable vs unstable, usually with examples. |
| #4 | Zero-force member add-on - Definition/purpose of zero-force members is occasionally attached to the classification question. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Classify five structures; draw SFD/BMD of a laterally and vertically loaded frame.
2017	Differentiate determinate/indeterminate and stable/unstable; classify structures; draw AFD/SFD/BMD of a frame.
2018	Purpose of zero-force members + classification; reactions and SFD/BMD of a frame.
2019	Classify structures; frame SFD/BMD using given member end moments.
2020*	Classification is visible in the shortened scan; the full companion diagram question is not fully available.
2021	Classify structures; overhang-beam SFD/BMD by area method appears in the same core block.
2022	Zero-force member + classification; frame reaction/SFD/BMD again appears in the core block.
2023	Classify determinacy/stability; draw SFD/BMD for a pitched rigid frame.
2024	Classification plus frame force-diagram problem.
2024 BL	Differentiate + classify; determine reactions and draw axial/shear/bending diagrams of a frame.

Practice from your uploaded files

- QS ANALYSIS- AYAN SIR (Stability + SFD/BMD pages)
- 1.0 Stability & Determinacy lecture
- 2.0 Shear Force and Bending Moment lecture
- Kabbo class note / Structure-Kabbo sir

Exam trap Do not classify from reaction count alone. Internal hinges, support directions and geometric instability can make a structure unstable even when a simple count looks acceptable.

SECTION A TRUSS + THREE-HINGED ARCH

| SET 2

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- Plane truss member forces by method of joints / method of sections / special truss layouts.
- Three-hinged parabolic arch: reactions, horizontal thrust, bending moment or N-Q-M at a specified point.
- Recent papers also ask conceptual beam-vs-arch comparisons and arch advantages.

Most repeated patterns - highest priority first

- | | |
|----|---|
| #1 | Three-hinged arch at a specified point - Find bending moment or internal forces at D/E or a specified distance from a support. This is extremely repetitive. |
| #2 | Selected truss member forces - Compute forces in named members and state tension/compression; method of joints is the dominant pattern. |
| #3 | Method of sections / full truss - Some years shift to method of sections or ask forces in all members under a UDL. |
| #4 | Arch theory add-on - Advantages of arch over beam, beam-vs-arch difference, maximum BM location, or pure-compression condition. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Method of joints for FG, GD, CB, CF; arch BM at a specified point.
2017	Selected truss member forces; three-hinged arch internal forces/moment at D.
2018	Method of sections for AB, BC, BI; arch BM at 30 m from support.
2019	Henneberg truss + typical truss shapes; arch BM at 30 m from support.
2020*	Arch BM at 25 m from support is visible in the shortened paper.
2021	Cantilever-truss member forces and a limiting-load truss; three-hinged arch pure-compression question.
2022	Method of joints; arch advantages + maximum BM location/magnitude.
2023	All-member truss under UDL; beam-vs-arch comparison + internal forces.
2024	Selected truss member forces; three-hinged parabolic arch internal-force/BM problem.
2024 BL	Truss FG/GD/CB/CF; arch internal forces and moment at D.

Practice from your uploaded files

- TRUSS QB + ASSIGNMENT MAHIR
- ARCH - QB SOLUTION - MAHIR
- 3.0 Analysis of Arches lecture
- Kabbo class note / Structure-Kabbo sir

Exam trap In arch problems, calculate the ordinate y of the arch at the section correctly before using $M_{\text{arch}} = M_{\text{beam}} - H \cdot y$. A small coordinate mistake propagates through N, Q and M.

SECTION A WIND + EARTHQUAKE

| SET 3

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- Calculate design wind load at each floor from building geometry, exposure/site information and coefficient tables.
- Calculate equivalent-static earthquake/seismic load at each floor from the supplied seismic parameters.
- Theory add-ons are less frequent than the numerical, but sustained/design pressure and vortex shedding have appeared.

Most repeated patterns - highest priority first

- | | |
|----|--|
| #1 | Floor-wise wind load - The most repeated wind template: compute wind pressure/force floor by floor from a plan/elevation and coefficient table. |
| #2 | Floor-wise seismic load - Equivalent Static Force Method with floor heights, dead load and seismic coefficients; distribute base shear to floors. |
| #3 | Recent code-style data - 2022-2024 increasingly provide location, risk/importance category, exposure/soil class and coefficient tables. |
| #4 | Theory add-on - Sustained vs design wind pressure, factors controlling wind magnitude, or vortex shedding. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Wind-pressure theory + 4-storey wind calculation; 4-storey earthquake load.
2017	5-storey wind; 6-storey hospital seismic load by equivalent static method.
2018	5-storey seismic + 5-storey wind.
2019	4-storey seismic + 4-storey wind.
2020*	4-storey wind and 5-storey seismic load are visible.
2021	4-storey wind; 5-storey steel-building seismic load.
2022	Vortex shedding + 7-storey wind; 5-storey earthquake load.
2023	7-storey wind (Mirpur) + 5-storey seismic load (Khulna) with site-specific tables.
2024	Wind + equivalent-static earthquake again in the core block.
2024 BL	5-storey wind + 5-storey earthquake, both full numerical problems.

Practice from your uploaded files

- 4.0 Analysis of Wind
- 5.0 Earthquake
- 0.0 Kabbo class note
- QS ANALYSIS- AYAN SIR (Wind & Earthquake pages)

Exam trap Use the coefficient/table values supplied in the paper. Do not mix older wind/seismic coefficients from another year with a recent problem.

SECTION A

| SET 4

DEFLECTION: UNIT LOAD / VIRTUAL WORK + CONJUGATE BEAM

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- Unit-load/virtual-work deflection or rotation of a beam, rigid frame or truss.
- Conjugate-beam method for slope and deflection at one or several points.
- Repeated theory/proof: define conjugate beam, real-vs-conjugate relations, and the unit-load deflection formula.

Most repeated patterns - highest priority first

- | | |
|----|--|
| #1 | Unit-load frame/beam deflection - Apply a unit load or unit moment at the target DOF; integrate mM/EI or sum member contributions. |
| #2 | Conjugate-beam numerical - Build the M/EI loading, convert supports, then obtain slope from conjugate shear and deflection from conjugate moment. |
| #3 | Conjugate-beam theory - Definition, characteristics and support/response relationship between real and conjugate beams. |
| #4 | Direct proof - Prove the unit-load formula $\Delta = \int (mM/EI) dx$; this exact derivation repeats. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Conjugate-beam definition; unit-load frame deflection; multi-output conjugate-beam numerical.
2017	Conjugate-beam relation; unit-load frame deflection; conjugate numerical.
2018	Prove unit-load deflection formula; truss-joint deflection by unit load.
2019	Frame rotation/horizontal deflection by virtual work; conjugate moments/max-deflection/slope.
2020*	Unit-load deflection proof is visible.
2021	Truss horizontal deflection by unit load; conjugate-beam moments/max deflection/slope.
2022	Frame rotation + horizontal deflection by virtual work; conjugate-beam relation + slope/deflection.
2023	Cantilever slope/deflection by unit load; conjugate slope/deflection at B and D.
2024	Frame joint rotation/deflection by virtual work; conjugate-beam definition + numerical.
2024 BL	Frame unit-load deflection + conjugate-beam definition/relationship and numerical.

Practice from your uploaded files

- 6.0 Deflection (Virtual Work)
- 7.0 Deflection (Conjugate Beam)
- Kabbo class note
- QS ANALYSIS- AYAN SIR (Deflection pages)

Exam trap For rotation, the virtual action is a unit moment; for translation, it is a unit force. In conjugate-beam problems, support conversion is just as important as the M/EI diagram.

SECTION B INFLUENCE LINE DIAGRAM (ILD)

| SET 5

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- ILD for vertical/horizontal reactions, support moments, shear and bending moment at specified points.
- Both beams and rigid frames with internal hinges appear in recent papers.
- Theory questions occasionally ask why ILDs are straight in determinate structures or their engineering importance.

Most repeated patterns - highest priority first

- | | |
|-----------|---|
| #1 | Beam ILD: reaction + shear + BM - The most repeated template. A unit load moves across a beam with overhangs/internal hinges and ordinates are required. |
| #2 | Frame ILD - Recent papers repeatedly include rigid frames and ask support reactions plus shear/BM at a point. |
| #3 | Internal-hinge / multi-support beam - Multiple support reactions and left/right shear at a support are common variations. |
| #4 | Theory - Why the ILD is straight for a statically determinate structure; importance/application of ILDs. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Two separate ILD questions: reactions/shear, plus reaction/shear/moment with internal hinges.
2017	Reactions at B, D, G and reaction moment at G.
2018	Beam ILD with ordinates; ILD importance appears with a mill-bent application.
2019	Frame ILD: reaction at A and shear/moment at specified points.
2020*	Beam ILD for reactions A/C/D and shear at points 1 and 2.
2021	Reactions at B/E/G + shear at C + moment at F.
2022	Detailed beam ILD ordinates and a separate frame ILD.
2023	Why ILD is straight in determinate structures + beam and frame ILD.
2024	Beam ILD + frame ILD with reactions/shear/BM.
2024 BL	Beam ILD + frame ILD again, confirming the recent pattern.

Practice from your uploaded files

- INFLUENCE LINE - QB SOLUTION
- Ramjan Sir ILD Assignment
- Ramjan class note / Structure-Ramjan sir

Exam trap Always define the moving-load position x and write piecewise equilibrium ranges around supports/hinges. Most ILD mistakes come from using one expression across a point where the structural condition changes.

SECTION B | SET 6

APPROXIMATE FRAME ANALYSIS: PORTAL + CANTILEVER + FACTOR

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- Portal method for lateral-load frames: axial force, shear, moments and complete member diagrams.
- Cantilever method using column areas/centroid and overturning moment.
- Factor method is historically important, but recent 2023-2024 papers emphasize portal + cantilever.

Most repeated patterns - highest priority first

- | | |
|-----------|--|
| #1 | Portal method - Find approximate axial forces, shears and moments in all members; sometimes draw full SFD/BMD. |
| #2 | Cantilever method - Use column-area ratios and the floor-group centroid; calculate column axial forces then shears/moments. |
| #3 | Assumptions - State assumptions of portal and/or cantilever method before the numerical. |
| #4 | Factor method - Repeated in 2017-2019; lower recent visibility but should remain a backup topic. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Portal/cantilever assumptions + full cantilever-method building-frame analysis.
2017	Cantilever assumptions/analysis; compare approximate methods + factor-method analysis.
2018	Cantilever frame + factor-method frame.
2019	Cantilever frame; portal assumptions + factor method.
2020*	The shortened scan is incomplete for this block.
2021	Portal/cantilever assumptions + cantilever frame; separate portal-method frame.
2022	Cantilever method + portal method paired in one question.
2023	Portal method + cantilever method paired in the recent standard form.
2024	Portal method + cantilever method paired again.
2024 BL	Portal method + cantilever method, same modern pattern.

Practice from your uploaded files

- Cantilever Method - QB SOLUTION - MAHIR
- Ramjan Sir Sheet (Full 3-1)
- Ramjan class note / Structure-Ramjan sir

Exam trap Do not mix assumptions: portal method distributes storey shear by column location, while cantilever method distributes column axial force according to distance from the centroid and column area.

SECTION B

| SET 7

MILL BENT / PORTAL APPLICATION + SPACE TRUSS

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- 3D space-truss reactions and member forces using three-dimensional equilibrium.
- Mill-bent or industrial-frame analysis, usually through portal-method assumptions and SFD/BMD of selected legs/girders.
- Space truss is the more stable annual core; mill-bent placement shifts between set positions in older papers.

Most repeated patterns - highest priority first

- | | |
|----|--|
| #1 | Space truss reactions + bar forces - The dominant pattern: elevation + plan are given; solve support reactions and member axial forces. |
| #2 | Space-truss theory - Definition, solution methods, determinacy and stability appear as short theory before the numerical. |
| #3 | Mill-bent SFD/BMD - Analyze selected legs/girders of a mill bent, usually with portal-method assumptions. |
| #4 | Recent pairing - 2023-2024 combine mill-bent/portal application with space truss in adjacent subparts. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Space-truss support reactions and member forces.
2017	Space truss again as a full numerical.
2018	Define space truss/methods + solve; separate mill-bent leg SFD/BMD.
2019	Space truss + mill-bent SFD/BMD.
2020*	The shortened scan is incomplete for this block.
2021	Full space-truss support-reaction/member-force problem.
2022	Define determinacy/stability of space truss + full 3D solution.
2023	Space truss + mill-bent portal-method application.
2024	Mill-bent portal application + space-truss numerical.
2024 BL	Mill-bent column diagrams + space-truss reactions/member forces.

Practice from your uploaded files

- Ramjan Sir Sheet (Full 3-1)
- Ramjan class note / Structure-Ramjan sir
- Structural Analysis and Design-I QB figures

Exam trap Resolve member directions in 3D before writing equilibrium. Use coordinates/direction cosines consistently; a wrong sign or swapped axis can ruin the entire space-truss solution.

SECTION B

| SET 8

THREE-MOMENT EQUATION + SUSPENSION CABLE

ANNUAL CORE

8/8 complete regular papers | 2024 backlog confirms

What this set usually contains

- Continuous-beam analysis by the three-moment equation, often including support settlement, overhangs or different I values.
- Suspension cable under UDL or point loads: reactions, horizontal component, maximum tension, dip and/or cable length.
- This is one of the most formulaic and persistent Section B families.

Most repeated patterns - highest priority first

- | | |
|----|--|
| #1 | Three-moment continuous beam - Find unknown support moments, reactions and usually SFD/BMD. Settlement and variable I recur often. |
| #2 | Cable maximum tension - Same-level or unequal-level supports; UDL or point loads; maximum tension is the most repeated output. |
| #3 | Cable length / dip - Many years add total cable length, specified dip, or position of the lowest point. |
| #4 | Special cable variants - Backstay/saddle tension or support-level difference appear as variations, but the equilibrium backbone stays the same. |

Year-wise evidence

Year	Past question pattern (paraphrased)
2016	Three-moment equation with settlement + diagram; cable max tension plus backstay/saddle tension.
2017	Three-moment continuous beam; suspension-bridge cable maximum tension.
2018	Three-moment beam with variable I; cable maximum tension + cable length.
2019	Three-moment with settlement/variable I; unequal-support cable max tension, dip and length.
2020*	Three-moment equation with support settlement is visible.
2021	Three-moment beam + unequal-level cable maximum tension.
2022	Three-moment beam + cable-supported roof under UDL.
2023	Three-moment beam + unequal-level UDL cable max tension.
2024	Three-moment beam + suspension cable max tension.
2024 BL	Three-moment beam with several settlements/variable I + unequal-level cable max tension.

Practice from your uploaded files

- 3 Moment Equation QB SOLUTION - MAHIR
- Cable Theory + Math
- Ramjan class note / Structure-Ramjan sir
- PRVS arif solve (arch/cable style supporting solutions)

Exam trap For three-moment problems, write span lengths, I values, load-resultant terms and settlements in one table before substitution. For cables, keep horizontal H separate from vertical reactions until the final tension calculation.

High-frequency Theory Question Bank

Priority rule Learn the first 8 questions as short, exam-ready answers. The remaining items are valuable backups and often appear as a 5-7 mark lead-in to a numerical.

#	Theory question	Priority
1	Differentiate statically determinate and indeterminate structures; differentiate stable and unstable structures, with examples.	Very High
2	Classify a group of beams/frames/trusses as stable/unstable and determinate/indeterminate.	Very High
3	What is a zero-force member? State its purpose / identify zero-force members.	High
4	State the advantages of arch construction over a beam / distinguish between a beam and an arch.	High
5	Define sustained wind pressure and design wind pressure. What factors control wind magnitude?	High
6	Explain vortex shedding effect.	Recent
7	Define a conjugate beam. State the basic characteristics/relationship between real and conjugate beams.	Very High
8	Why is an influence line a straight line for a statically determinate structure?	Recent
9	State the importance/applications of influence line diagrams in building/structural engineering.	High
10	Define space truss. State methods of solution and discuss determinacy/stability.	High
11	Write the assumptions of the portal method and cantilever method.	Very High
12	Which approximate method is more accurate for a laterally loaded frame? Give reasons.	Medium
13	Draw/identify typical Fink, saw-tooth and crescent trusses.	Medium
14	State the sign/response relations used in the conjugate-beam method (shear $\rightarrow$ slope, moment $\rightarrow$ deflection).	High

Theory answers that should include a sketch

- Stability/determinacy classification when the support/hinge arrangement is central to the explanation.
- Beam vs arch comparison.
- Typical truss shapes.
- Conjugate-beam support conversion / real-conjugate relationship.
- Portal vs cantilever method assumptions (neutral axis / inflection points).

Proof / Derivation / Formula-Logic List

Directly repeated proof The only clearly repeated standalone proof in the provided QB is the unit-load/virtual-work deflection relation. The other items below are recurring derivation logic needed inside numericals rather than repeatedly asked as full proofs.

A. Direct proof - MUST KNOW

Prove the beam deflection relation by unit-load method: $\Delta = \int_0^L \frac{mM}{EI} dx$.

QB evidence: Asked directly in 2018 and again in the shortened 2020 paper.

B. Conjugate-beam mapping

Real-beam M/EI diagram becomes loading on the conjugate beam; conjugate shear corresponds to real slope and conjugate moment to real deflection.

QB evidence: Repeated as theory + numerical across many years.

C. Three-hinged parabolic arch equation

Use $y = 4h x (L-x) / L^2$ and $M_{\text{arch}} = M_{\text{beam}} - Hy$; resolve N and Q using the local tangent angle when requested.

QB evidence: Embedded in nearly every arch numerical.

D. Suspension-cable UDL relation

For same-level supports under UDL, the cable is parabolic and the horizontal component is tied to span, load and dip; then combine H with vertical reactions for maximum tension.

QB evidence: Embedded in annual cable numericals.

E. Three-moment general form

Know the full three-moment equation including unequal I and support-settlement terms; be able to reduce it for constant EI /no settlement cases.

QB evidence: Settlement and varying I recur frequently.

F. Influence-line piecewise equilibrium

Derive ordinates by moving a unit load and writing equilibrium over each x -range separated by supports/internal hinges.

QB evidence: Core derivation logic for every ILD problem.

Recent Trend: what changed in 2022-2024?

Wind & earthquake	More site/location/exposure/soil/risk-category data and supplied coefficient tables. Practice the exact table-driven workflow, not only old examples.
ILD	Recent papers use both beams and frames, often with internal hinges; 2023 adds a short conceptual ILD question.
Approximate analysis	Portal + cantilever dominate 2022-2024. Factor method remains historically repeated but is less visible in the latest papers.
Set 7	Mill-bent portal applications and space-truss numericals appear together/adjacent in 2023-2024.
Three-moment + cable	Still highly stable: continuous beam + cable problem remains a reliable recurring family.
Deflection	Unit load/virtual work + conjugate beam remains a dependable two-method pair.
2024 Backlog	The same eight study families reappear, but the printed Section A/B blocks are reversed compared with the 2024 Regular paper.

If you have limited time, solve these first

1. One recent wind + earthquake pair (2023/2024 style).
2. One unit-load frame + one conjugate-beam question.
3. One three-hinged arch point-force question + one selected-member truss.
4. One classification problem + one complete frame SFD/BMD.
5. One beam ILD + one frame ILD.
6. One portal-method frame + one cantilever-method frame.
7. One complete space-truss problem.
8. One three-moment problem with settlement/variable I + one unequal-level cable problem.

Final 3-of-4 Exam Checklist

SECTION A Choose any 3 sets you can execute completely

- Set 1: Can I classify hinges/supports correctly and draw complete SFD/BMD without sign confusion?
- Set 2: Can I solve a truss systematically and compute arch y , H , M , N and Q at a section?
- Set 3: Can I finish floor-wise wind and earthquake load tables without mixing coefficients?
- Set 4: Can I set up unit-load mM/EI and convert real supports to conjugate supports correctly?

SECTION B Choose any 3 sets you can execute completely

- Set 5: Can I write piecewise ILD ordinates for reactions, shear and moment in beams/frames?
- Set 6: Can I state portal/cantilever assumptions and complete the approximate force distribution?
- Set 7: Can I solve 3D direction cosines/equilibrium for a space truss and handle mill-bent diagrams?
- Set 8: Can I write the correct three-moment equation (including settlement/ I) and solve cable H , reactions and maximum tension?

Minimum safe preparation target Be exam-ready in 6 sets total: 3 strong sets in Section A + 3 strong sets in Section B. Then use the remaining two sets only as backups. Because every set family is annual-core in the complete papers reviewed, completely abandoning a section family is riskier than keeping a compact backup method sheet.

Source corpus used

- Structural Analysis and Design - I QB (2016-2024)
- 2024 Backlog Exam
- QS ANALYSIS - AYAN SIR
- Kabbo and Ramjan class notes / topic lecture slides
- Topic-wise solved QB files: Truss, Arch, Influence Line, Three-Moment, Cantilever, Cable/supporting solutions

Made by 2223033

CE 3111 - Structural Analysis and Design - I